



بسم الله الرحمن الرحيم

Course Name:
CS251 Intro to Software Engineering
Course Instructor:
Dr. Mohammad El-Ramly

Team Details:

ID	Name	Email	Mobile
20240394	Amr Tamer Abdellatif	20240394@stud.fci-cu.edu.eg	01097225047
20240093	Andro Hazem Samir	20240093@stud.fci-cu.edu.eg	01274681549
20240150	Hazem Hosny Hamed	20240150@stud.fci-cu.edu.eg	01099516007
20242450	Manal Ezzat Ahmed	20242450@stud.fci-cu.edu.eg	01100427505

A.software engineers Jobs

1- Full-Stack Developer Role at Elsewedy Electric PSP:

- Elsewedy Electric PSP is looking for a driven Full-Stack Developer to work across the entire stack, building back-end services with .NET, front-end experiences with Angular for enterprise-grade applications.

Key Responsibilities:

The role covers a wide range of engineering tasks, including:

- Building and maintaining RESTful APIs and back-end services (ASP.NET Core)
- Developing responsive, modular front-end features (Angular and TypeScript)
- Debugging and resolving issues across front-end, API, and database layers
- Writing clean, maintainable, and well-documented code • Working collaboratively within an Agile/Scrum team

Requirements & Qualifications:

- **Back-end:** Solid understanding of C#, .NET / ASP.NET Core, SQL Server, OOP, and SOLID principles
- **Front-end:** Working knowledge of Angular (v12+), TypeScript, RxJS, HTML, and CSS
- **Education:** Bachelor's degree in Computer Science, Software Engineering, or a related field
- **Soft Skills:** Fast learner with a proactive attitude, good communication skills, and a strong commitment to scalable code
- **Nice to Have:** Hands-on experience with Git, Azure services, state management (NgRx), CI/CD, and unit testing

Application Link:

[Full Stack Developer | ELSEWEDY ELECTRIC POWER SYSTEMS PROJECTS | LinkedIn](#)

- Freelance:

As a freelance Full-Stack Developer, instead of working full-time for one company, you operate as an independent business. You offer your technical skills to various clients on a project-by-project or hourly basis.

Types of Freelance Jobs You Can Offer:

Custom Web Application Development: Clients often need internal dashboards, CRM systems, or SaaS (Software as a Service) products. You would build the secure database and logic using ASP.NET Core and create the interactive user interface using Angular.

API Development & Integration: You can offer services to build backend RESTful APIs using .NET that a client might need to connect their existing mobile apps, front-end websites, or third-party tools.

Bug Fixing and Maintenance: Companies sometimes hire freelancers just to resolve specific bugs, optimize database queries.

2-Jobs Ai

The **Software Engineer** role is a fully remote position with a competitive pay range of **\$40–\$150 per hour**. The company is seeking engineers who can design, develop, and deliver production-level software solutions using modern programming languages such as **Python, TypeScript, JavaScript, Go, Rust, and Java**.

Required Qualifications:

- At least **1 year of software engineering experience**
- Strong knowledge of one or more programming languages
- Good understanding of **data structures, algorithms, APIs, and databases**
- Experience with **Git, testing, and code review processes**
- Strong written English and communication skills
- Ability to work independently in a remote environment

Preferred Skills:

- Experience with **AWS, GCP, or Azure**
- Knowledge of **Docker, Kubernetes, and CI/CD**
- Familiarity with **AI tools, LLMs, or RAG pipelines**
- Open-source projects or portfolio work
- Experience with microservices systems

Application Link: <https://www.linkedin.com/jobs/view/4404048096>

3- A Software Tester:

➡ is the "safety net" of the development process. Their job is to try and break a piece of software before a customer uses it. They identify bugs, performance issues, and usability flaws to ensure the final product is stable and user-friendly

1. What exactly does a Software Tester do?

The role is divided into two main categories:

a. Manual Testing

- What it is: You act like a real user, clicking buttons and exploring features to find errors.
- Tasks: Writing test cases (step-by-step instructions), reporting bugs in tools like Jira, and verifying that a bug was actually fixed.

b. Automation Testing

- What it is: You write scripts (using code like Python or Java) that test the software automatically.
- Tasks: Writing scripts in tools like Selenium or Cypress to run repetitive tests very quickly.

2. How to do it as a Freelancer

Freelancing in testing is unique because you don't always need a long-term contract; many companies pay for "crowdsourced" testing or short-term audits.

➤ Start with "Crowd-testing" Platforms

This is the easiest way to start without an interview. These sites pay you per bug found.

- uTest: The largest community. They have a "uTest Academy" to teach you for free, and then they invite you to paid cycles

<https://www.utest.com/#:~:text=uTest%20is%20the%20home%20of,digital%20experiences%20to%20their%20customers.>

- Test IO: You get paid for every bug you find in real-world apps.
- UserTesting: Focuses on Usability Testing. You record your screen and voice while using an app and talk about your experience.

4- AI Engineer

Job Description

Are you ready to redefine the boundaries of what's possible with Artificial Intelligence? We are searching for an exceptional **AI Engineer** to join our visionary team in the vibrant city of **Giza**,

Egypt. This is an unparalleled opportunity to immerse yourself in a culture of innovation, tackle complex challenges, and contribute to groundbreaking AI solutions that will shape the future. If you are passionate about pushing technological frontiers and making a tangible impact, this role is for you!

Overview:

As an AI Engineer, you will be at the forefront of designing, developing, and deploying cutting-edge intelligent systems. You will work on dynamic projects, from conceptualization to execution, bringing innovative ideas to life. This **on-site** role offers a collaborative environment where your expertise will be pivotal in delivering high-impact AI products and services.

Responsibilities:

- **Design and implement advanced AI/ML models** and algorithms, ensuring robust and scalable solutions.
- Conduct thorough research into new AI technologies and methodologies to continuously enhance our capabilities.
- **Collaborate closely with cross-functional teams**, including software developers, data scientists, and product managers, to integrate AI solutions seamlessly into existing systems.
- Develop and optimize data pipelines, perform data analysis, and pre-process large datasets to prepare them for model training.
- **Evaluate, test, and fine-tune AI models** for performance, accuracy, and efficiency.
- Stay abreast of industry trends, emerging technologies, and best practices in AI and machine learning.

Skills & Technologies:

You will leverage a powerful toolkit including **AI, Machine Learning, and Deep Learning** techniques. Proficiency in programming languages like **Python**, coupled with experience in frameworks such as **TensorFlow** and **PyTorch**, will be essential. A strong foundation in **Computer Science, Information Technology (IT), Data Science, and Natural Language Processing** will be your keys to success.

Growth Opportunities:

We are deeply invested in your professional development. You'll gain exposure to diverse and challenging projects, enabling continuous skill enhancement and career progression. Expect opportunities for mentorship, advanced training, and the chance to lead initiatives as you grow with us.

Job Requirements

- **Academic Excellence:** A strong academic foundation in **Computer Science, AI, Information Technology (IT)**, or a closely related technical field is highly valued.
- **Proven Experience:** A solid track record of **1-10 years of professional experience** in AI development, machine learning engineering, or a related field.
- **Core AI Expertise:** Demonstrated hands-on experience with **AI, Machine Learning**, and **Deep Learning** algorithms and methodologies.
- **Programming Prowess:** Exceptional proficiency in **Python** and experience with key AI/ML libraries and frameworks such as **TensorFlow** and **PyTorch**.
- **Data Savvy:** Strong understanding and practical experience in **Data Science**, data manipulation, and pre-processing techniques.
- **Problem-Solving Mastery:** Excellent analytical and **Problem Solving** skills, with the ability to design innovative solutions to complex challenges.
- **NLP Skills:** Familiarity with **Natural Language Processing** concepts and applications is a significant advantage.
- **Collaboration & Communication:** Ability to work effectively in a team environment and communicate complex technical concepts clearly.

Job page link:

<https://wuzzuf.net/jobs/p/ixh5hukifyyc-ai-engineer-giza-egypt?o=1&l=sp&t=sj&a=AI|search-v3|navbg>

B. Government Training & National Initiatives (Egypt)

1- Backend Engineering with Java, Kotlin, and Modern Architectures Program:

- The National Telecommunication Institute (NTI) offers this comprehensive training program (as part of the Banq Masr initiative) designed to provide foundational and advanced skills for building secure, scalable, and maintainable enterprise-grade backend systems.

Training Fields:

The program covers a wide range of advanced engineering areas, including:

- Java Essentials, Object-Oriented Programming (OOP), and Concurrency
- Kotlin for Backend and Spring Boot Framework
- API Design and Security (JWT, OAuth2)
- Microservices Architecture, Docker, and Kubernetes
- SQL Databases (Oracle/PostgreSQL) and Performance Optimization
- Workflow Automation and BPMN Models with Camunda

Key Benefits & Requirements:

- Suitable for software engineers, CS graduates, and Java developers transitioning to Kotlin/Spring Boot
- Extensive learning path consisting of 210 Hours (50+ Days)
- Convenient schedule: Training is held after official working hours (starting at 6:00 PM for 4 hours daily)
- Pre-requisites include basic programming knowledge (OOP, Data Structures, SQL, and client-server architecture)

Application Link: <https://www.nti.sci.eg/bmasr/>

2-ITI Academy for Undergraduates

ITI Academy for Undergraduates is a training platform provided by the Information Technology Institute (ITI) in Egypt. It is designed especially for university students to help them improve their technical and professional skills before graduation. The academy offers camps, workshops, short courses, and career talks during summer and mid-year vacations. Programs cover fields such as web development, software development, digital arts, networks, cybersecurity, and information systems.

Application Link: <https://academy.iti.gov.eg/>

3-Digital Egypt Pioneers (رواد مصر الرقمية):

It is one of the most important current government initiatives in Egypt, launched by the Ministry of Communications and Information Technology (MCIT). The initiative aims to prepare a generation of qualified professionals in the latest technological fields to enable them to compete in the global job market and pursue freelance work.

1. Technology Tracks

The initiative offers 7 specialized tracks designed to match global market demands:

- **Software Development:** Web and mobile application programming.
- **AI & Data Science:** Machine learning and big data analytics.
- **Cybersecurity:** Protecting networks and data from digital attacks.
- **Embedded Systems:** Programming hardware components (crucial for automotive/IoT).
- **Digital Arts:** Graphic design, 3D modeling, and motion graphics.
- **Business Analysis:** Bridging the gap between IT and business requirements.

- **ERP (Enterprise Resource Planning):** Managing business processes via systems like SAP or Oracle.

2. How to Apply

1. **Visit the Official Website:** Go to depi.gov.eg.
2. **Create an Account:** Register using your National ID and a valid email.
3. **Choose Your Category:** Select whether you are applying as a **"Student"** or a **"Graduate."**
4. **Select Your Track:** Pick one of the 7 tracks listed above that fits your career goals.
5. **Upload Required Documents:** Usually requires a National ID copy, proof of university enrollment (for students), or a graduation certificate (for graduates).
6. **Take the Assessment:** You will be invited to take online entrance exams (English and Logic/Basic Technical skills).
7. **Interview:** Successful candidates from the exams may be called for a brief technical or personal interview.

4- NTI Train To Hire (4 Month)

The initiative is offered as a full-fledged scholarship for selected fresh Egyptian university graduates within five calendar years of their graduation. It helps in bridging the gap between the talent skills and the regional and international market need. Collaboration with more than 75 national and multinational industry partners and more than 30 professional tracks. Candidates will join after a fine screening process done in cooperation with our industry partners. Tracks are designed in cooperation with our industry partners to meet job profiles and ICT market needs. Each track has a duration of four months including technical program, soft skills program, in addition to a one month on job training at the industry partner premises. After passing the track the candidates will receive joint certificates from NTI and our industry partners.

The initiative offers a full-fledged scholarship and targets 2500 fresh graduates from Egyptian universities nationwide. The initiative adopts the upskilling approach: technical up-skilling and personal and soft skills up-skilling aspects. The program seeks to cover various digital skills to cater to diverse interests and career paths.

Tracks:

- Electronics and Embedded Systems
- Network Infrastructure and Enablers

- Cybersecurity
- DevOps Engineering
- Telecom Engineering
- Software Engineering

Initiative link: <https://www.nti.sci.eg/dey/HireReady.html>

C.Internships

1-Software Engineering Intern, PhD Role at Google (Summer 2026):

- Google is looking for students to join their engineering team for a 12-14 week paid summer internship. In this role, you will work on complex computer science solutions, develop scalable, distributed software systems, and contribute to projects critical to Google's needs across search, cloud, AI, and infrastructure.

Key Responsibilities:

This internship focuses on high-level research and application, including:

- Researching, conceiving, and developing software applications to improve Google's product offerings.
- Working on massive scalability, storage solutions, and large-scale applications.
- Contributing to projects utilizing artificial intelligence, machine learning, data mining, compilers, or natural language processing.
- Designing, testing, deploying, and maintaining software while collaborating with peers and senior engineers to push technology forward.

Requirements & Qualifications:

- **Education:** Currently enrolled in a PhD degree program with a focus on Computer Science, Software Development, or a related technical field.
- **Technical Expertise:** Deep experience in data structures, algorithms, complexity analysis, and advanced software design.
- **Programming:** Strong coding experience in one or more general-purpose languages such as C/C++, Java, Python, or Go.
- **Specialized Knowledge:** Research experience in areas like distributed systems, computer architecture, machine learning, or networking is highly preferred.

- **Availability:** Must be able to work full-time (40 hours/week) for 12-14 weeks during the Summer of 2026 and return to the degree program afterward.

Application Link: [Software Engineering Intern, PhD, Summer 2026 — Google Careers](#)

2- Blink22 Software Engineering Intern

The Software Engineer Intern position at Blink22 is a remote full-time internship based in Alexandria, Egypt. The company is looking for motivated students who are passionate about software development and eager to work in a collaborative environment.

Required Qualifications:

- Currently studying **Computer Science, Engineering, or a related field**
- Basic knowledge of **Data Structures, Object-Oriented Programming (OOP), and Design Patterns**
- Familiarity with **web technologies**
- Ability to learn new tools and technologies quickly

Required Soft Skills:

- Strong communication skills
- Problem-solving and analytical thinking
- Ability to work independently and in teams
- Positive attitude and willingness to learn continuously
- **Application Link:** <https://www.linkedin.com/jobs/view/4377830329>

3-Backend Intern :

This internship is a Backend Developer position at Money Fellows, a prominent Egyptian fintech company. Unlike the previous "heavyweight" role you shared, this one is a true entry-level learning opportunity designed for students or recent graduates.

Here is a brief breakdown:

The Core Goal

The role focuses on helping you transition from academic knowledge to professional software engineering. You will be working on the "engine room" of their app, which serves over 7 million users.

Key Requirements

- Education: Currently studying or recently graduated in CS or Software Engineering.
- Tech Stack: Strong basics in C# (or similar OOP languages like Java/C++).
- Fundamentals: You should understand how Databases work, what a REST API is, and how to use Git.
- Soft Skills: They prioritize a "problem-solving mindset" and the ability to work well within a team.

What You Will Actually Do

- Guided Development: You won't be left alone; you'll build APIs and backend services under the guidance of senior engineers.
- Maintenance: You will help fix bugs and add small features to existing systems.
- Collaboration: You'll participate in Code Reviews, which is where you learn "industry-standard" coding practices.
- Documentation: You'll learn to document your code and technical findings properly.

Why It's a Good Opportunity

- Mentorship: The job description explicitly mentions "under guidance" and "learning," making it a safe space to grow.
- Impact: You are working on a product with millions of users, which looks excellent on a CV.
- Career Path: It's labeled as an "Early Career Program," often a stepping stone to a full-time paid position within the company.

<https://www.linkedin.com/jobs/view/4404998177/>

4- Software Engineering Intern - Avidbeam Technologies

Job Description:

A Software Engineer Intern assists in developing, testing, and debugging software applications under the guidance of experienced engineers. Interns typically work on real-world projects, often involving a "starter project" to learn the codebase and workflow, contributing to full-cycle development.

The intern will learn:

- Coding & Development

- Debugging & Testing: Identify, troubleshoot, and fix software defects.
- Collaboration: Participate in team meetings, code reviews, and pair programming sessions.
- Documentation: Document technical specifications, project progress, and API usage.
- Feature Implementation: Assist in building, testing, and deploying new features or software components.

Internship Requirements

- Senior Graduates in Computer Science, Software Engineering, or a related field.
- Basic understanding of software development principles, data structures, and algorithms.
- Proficiency in programming language.
- Strong analytical, problem-solving, and communication skills

Application Deadline: June 15th 2026

Internship Start date: July 1st 2026

Duration: 2 Months

Intern link:

<https://wuzzuf.net/internship/luimrbjskiw-software-engineer-intern-avidbeam-technologies-cairo-egypt?o=1&l=sp&t=sj&a=AI%20intern|search-v3|navbg>



1- Jobs Ai

A fully remote Software Engineer opportunity with a pay range of \$40–\$150 per hour. The role focuses on developing production-level software using technologies such as Python, JavaScript, TypeScript, Go, Rust, and Java. Suitable for candidates with at least one year of experience and strong problem-solving skills.

Application Link: <https://www.linkedin.com/jobs/view/4404048096>

2- ITI Academy for Undergraduates

ITI Academy for Undergraduates is a training platform provided by the Information Technology Institute in Egypt. It offers camps, workshops, and short courses for university students in areas such as web development, cybersecurity, networks, and software development. It helps students gain practical skills before graduation.

Application Link: <https://academy.iti.gov.eg/>

3- Blink22 Software Engineering Intern

A remote full-time internship opportunity at Blink22 in Alexandria, Egypt. The internship is designed for students in Computer Science or Engineering who have basic programming knowledge and want to gain practical experience in software development.

Application Link: <https://www.linkedin.com/jobs/view/4377830329>



1- A Software Tester:

➡ is the "safety net" of the development process. Their job is to try and break a piece of software before a customer uses it. They identify bugs, performance issues, and usability flaws to ensure the final product is stable and user-friendly

1. What exactly does a Software Tester do?

The role is divided into two main categories:

a. Manual Testing

What it is: You act like a real user, clicking buttons and exploring features to find errors.

Tasks: Writing test cases (step-by-step instructions), reporting bugs in tools like Jira, and verifying that a bug was actually fixed.

b. Automation Testing

What it is: You write scripts (using code like Python or Java) that test the software automatically.

Tasks: Writing scripts in tools like Selenium or Cypress to run repetitive tests very quickly.

2. How to do it as a Freelancer

Freelancing in testing is unique because you don't always need a long-term contract; many companies pay for "crowdsourced" testing or short-term audits.

➤ Start with "Crowd-testing" Platforms

This is the easiest way to start without an interview. These sites pay you per bug found.

uTest: The largest community. They have a "uTest Academy" to teach you for free, and then they invite you to paid cycles

<https://www.utest.com/#:~:text=uTest%20is%20the%20home%20of,digital%20experiences%20to%20their%20customers.>

Test IO: You get paid for every bug you find in real-world apps.

1- AI Engineer Job Requirements

- Academic Excellence: A strong academic foundation in Computer Science, AI, Information Technology (IT), or a closely related technical field is highly valued.
- Proven Experience: A solid track record of 1-10 years of professional experience in AI development, machine learning engineering, or a related field.
- Core AI Expertise: Demonstrated hands-on experience with AI, Machine Learning, and Deep Learning algorithms and methodologies.
- Programming Prowess: Exceptional proficiency in Python and experience with key AI/ML libraries and frameworks such as TensorFlow and PyTorch.
- Data Savvy: Strong understanding and practical experience in Data Science, data manipulation, and pre-processing techniques.
- Problem-Solving Mastery: Excellent analytical and Problem Solving skills, with the ability to design innovative solutions to complex challenges.
- NLP Skills: Familiarity with Natural Language Processing concepts and applications is a significant advantage.
- Collaboration & Communication: Ability to work effectively in a team environment and communicate complex technical concepts clearly.

Job page link: <https://wuzzuf.net/jobs/p/ixh5hukifyyc-ai-engineer-giza-egypt?o=1&l=sp&t=sj&a=AIsearch-v3|navbg>

2- NTI Train To Hire (4 Month)

The initiative offers a full-fledged scholarship and targets 2500 fresh graduates from Egyptian universities nationwide. The initiative adopts the upskilling approach: technical up-skilling and personal and soft skills up-skilling aspects. The program seeks to cover various digital skills to cater to diverse interests and career paths.

Tracks:

- Electronics and Embedded Systems
- Network Infrastructure and Enablers
- Cybersecurity
- DevOps Engineering
- Telecom Engineering
- Software Engineering

Initiative link: <https://www.nti.sci.eg/dey/HireReady.html>

3- Software Engineering Intern - Avidbeam Technologies

The intern will learn:

- Coding & Development
- Debugging & Testing: Identify, troubleshoot, and fix software defects.
- Collaboration: Participate in team meetings, code reviews, and pair programming sessions.
- Documentation: Document technical specifications, project progress, and API usage.



STUD 20240394 • 8:45 PM

1- Full-Stack Developer Role at Elsewedy Electric PSP:

- Elsewedy Electric PSP is looking for a driven Full-Stack Developer to work across the entire stack, building back-end services with .NET, front-end experiences with Angular for enterprise-grade applications.

Key Responsibilities:

The role covers a wide range of engineering tasks, including:

- Building and maintaining RESTful APIs and back-end services (ASP.NET Core)
- Developing responsive, modular front-end features (Angular and TypeScript)
- Debugging and resolving issues across front-end, API, and database layers
- Writing clean, maintainable, and well-documented code • Working collaboratively within an Agile/Scrum team

Requirements & Qualifications:

- Back-end: Solid understanding of C#, .NET / ASP.NET Core, SQL Server, OOP, and SOLID principles
- Front-end: Working knowledge of Angular (v12+), TypeScript, HTML, and CSS
- Education: Bachelor's degree in Computer Science, Software Engineering, or a related field
- Soft Skills: Fast learner with a proactive attitude, good communication skills, and a strong commitment to scalable code
- Nice to Have: Hands-on experience with Git, Azure services, state management, CI/CD, and unit testing

Application Link:

Full Stack Developer | ELSEWEDY ELECTRIC POWER SYSTEMS PROJECTS | LinkedIn